

4DVarNet-FM: Training Experiments (6 runs, S*)

Experiment Hyperparameters

ID	Model	Channels	S1Ep	S2Ep	TimeEmb	N_outer	SigPrior	TrainMix	Rand	ParamNoise
S1_direct_unet_s0s1	DirectUNet	[64, 128, 256]	400	0	-	-	-	s0_s1	no	-
S2_direct_unet_s0s1_small	DirectUNet	[32, 64, 128]	300	0	-	-	-	s0_s1	no	-
S3_vanilla_cfm_s0s1	VanillaCFM	[64, 128, 256]	600	0	64	10	0.5	s0_s1	no	-
S4_vanilla_cfm_s0s1_small	VanillaCFM	[32, 64, 128]	600	0	64	10	0.5	s0_s1	no	-
S5_joint_cfm_s0s1	JointCFM	[64, 128, 256]	600	0	64	10	0.5	s0_s1	no	-
S6_joint_cfm_s0s1_tau0	JointCFM	[64, 128, 256]	600	0	64	10	0.5	s0_s1	no	-

Shared settings (config/lorenz63_default.yaml):
dt=0.01, T_max=3.0s, obs_interval=20, R_var=0.5, B_var=2.0
num_windows=2000, spinup_steps=10000, batch_size=32, dropout=0.1
optimizer=Adam, gradient_weight=0.1 (grad-loss term)
Stage 2 only ever runs for Tweedie (S2Ep>0 above); DirectUNet/VanillaCFM are stage-1-only.

S* (data_setup=s0_s1): train/val use randomized param_bias in [0, bias_max] and forcing_state_bias=0 (RandomBiasLorenz63Dataset); eval on two held-out test sets:
S0: param_bias=0.0, forcing_state_bias=0.0 (in-distribution, no bias)
S1: param_bias=0.15, forcing_state_bias=0.1, fixed (out-of-distribution bias)

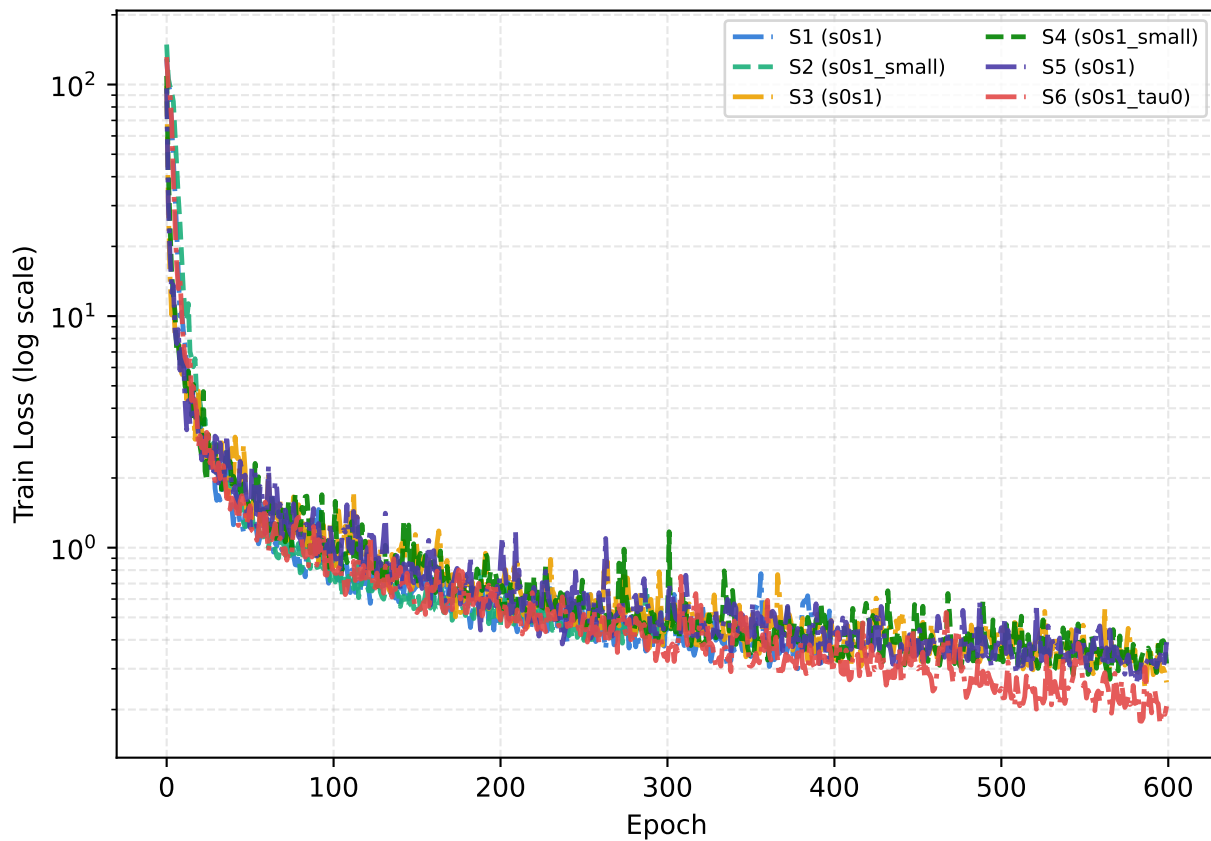
DirectUNet: single UNet1D pass, obs -> state (MSE + gradient loss), no flow matching.
VanillaCFM: conditional flow matching, LinearInterpolant(nu=1.0), Euler sampling over N_outer=10 steps, x0 ~ N(0, sigma_prior).
JointCFM: VanillaCFM backbone jointly conditioned on/predicting model parameters (param_dim, param_loss_weight); train_tau_0 only restricts training to tau=0.
Tweedie: 2-stage solver -- stage1 fits a Gaussian mean estimator, stage2 fits a non-Gaussian residual correction on top of the frozen stage1 estimator.

Run Status

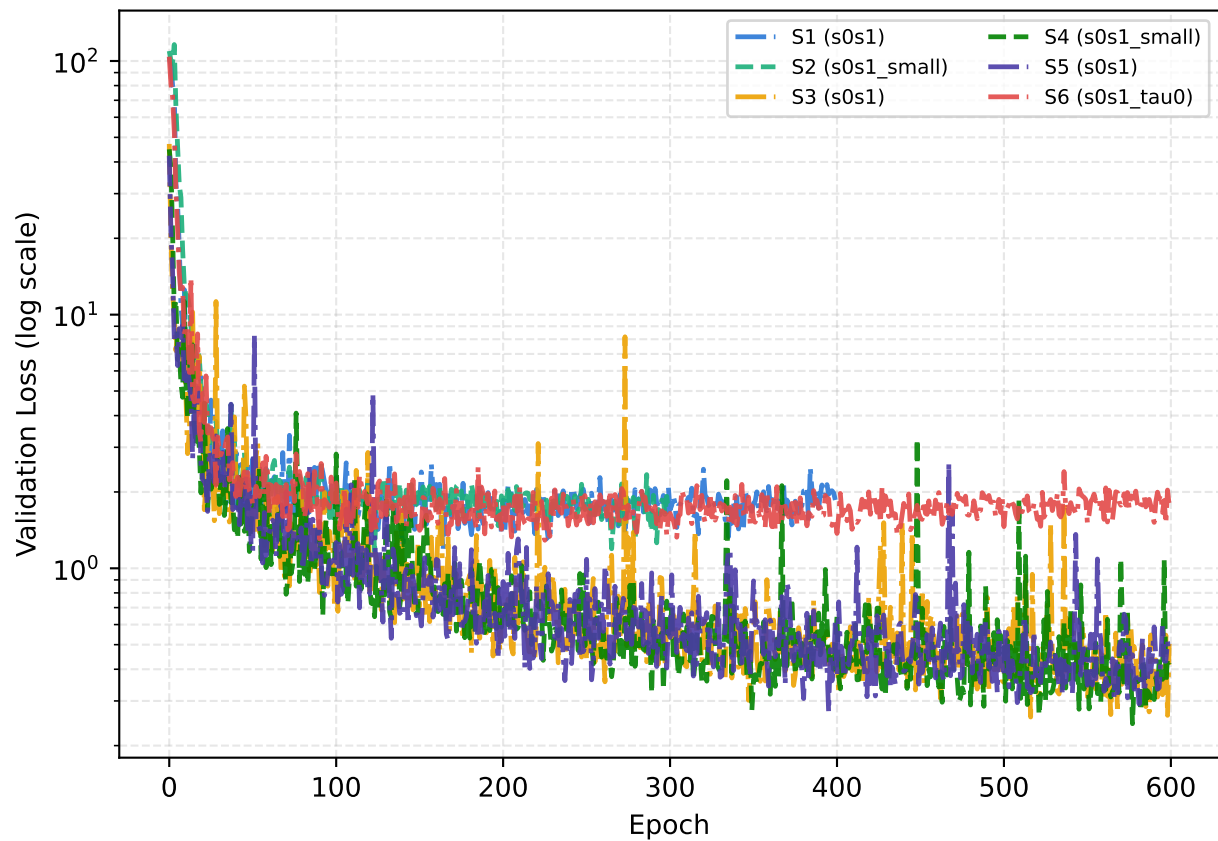
S1_direct_unet_s0s1	trained (results + loss log)
S2_direct_unet_s0s1_small	trained (results + loss log)
S3_vanilla_cfm_s0s1	trained (results + loss log)
S4_vanilla_cfm_s0s1_small	trained (results + loss log)
S5_joint_cfm_s0s1	trained (results + loss log)
S6_joint_cfm_s0s1_tau0	trained (results + loss log)

Training & Validation Loss

Train Loss



Validation Loss



Results Summary (S0 / S1)

Per-Variable Mean RMSE Summary (S0 / S1)

--- S0 ---

ID	Model	Trained On	X	Y	Z	Mean
S1_direct_unet_s0s1	DirectUNet	s0_s1	0.3465	0.5406	0.6981	0.5284
S2_direct_unet_s0s1_small	DirectUNet	s0_s1	0.3911	0.6058	0.7568	0.5846
S3_vanilla_cfm_s0s1	VanillaCFM	s0_s1	0.7168	0.9345	1.0211	0.8908
S4_vanilla_cfm_s0s1_small	VanillaCFM	s0_s1	0.5880	0.8902	0.9406	0.8063
S5_joint_cfm_s0s1	JointCFM	s0_s1	0.6170	0.8392	0.9663	0.8075
S6_joint_cfm_s0s1_tau0	JointCFM	s0_s1	0.3862	0.6499	0.7564	0.5975

--- S1 ---

ID	Model	Trained On	X	Y	Z	Mean
S1_direct_unet_s0s1	DirectUNet	s0_s1	0.6965	0.8829	2.4686	1.3493
S2_direct_unet_s0s1_small	DirectUNet	s0_s1	0.7451	1.1129	2.3155	1.3912
S3_vanilla_cfm_s0s1	VanillaCFM	s0_s1	0.7905	1.1021	2.2895	1.3940
S4_vanilla_cfm_s0s1_small	VanillaCFM	s0_s1	0.7103	1.2503	2.7948	1.5851
S5_joint_cfm_s0s1	JointCFM	s0_s1	0.7627	1.1681	2.5858	1.5055
S6_joint_cfm_s0s1_tau0	JointCFM	s0_s1	0.7104	1.0098	2.7728	1.4977

Degradation (S1 mean / S0 mean) and training time:

ID	Model	Trained On	S0 mu	S1 mu	Deg	Time(s)
S1_direct_unet_s0s1	DirectUNet	s0_s1	0.5284	1.3493	2.55	x 709
S2_direct_unet_s0s1_small	DirectUNet	s0_s1	0.5846	1.3912	2.38	x 543
S3_vanilla_cfm_s0s1	VanillaCFM	s0_s1	0.8908	1.3940	1.56	x 1138
S4_vanilla_cfm_s0s1_small	VanillaCFM	s0_s1	0.8063	1.5851	1.97	x 1116
S5_joint_cfm_s0s1	JointCFM	s0_s1	0.8075	1.5055	1.86	x 1211
S6_joint_cfm_s0s1_tau0	JointCFM	s0_s1	0.5975	1.4977	2.51	x 1189

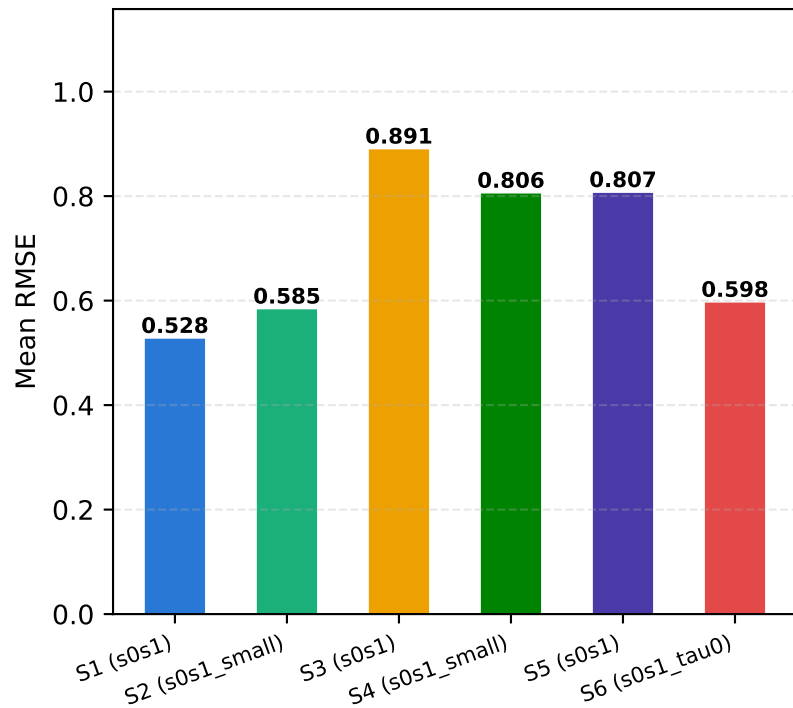
Parameter RMSE Summary (S0 / S1) — JointCFM

Parameter RMSE Summary (S0 / S1) – JointCFM experiments				
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--- S0 ---				
ID	sigma	rho	beta	c1

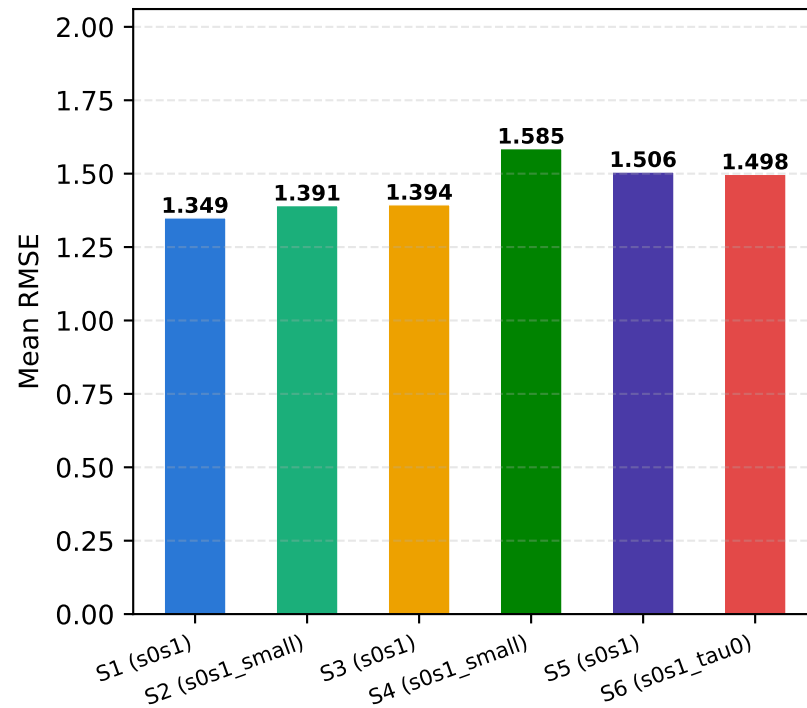
S5_joint_cfm_s0s1	0.2464	0.2778	0.3160	0.0322
S6_joint_cfm_s0s1_tau0	8.6677	28.3656	4.1640	0.9173
--- S1 ---				
ID	sigma	rho	beta	c1

S5_joint_cfm_s0s1	1.4082	3.6498	0.1037	0.0416
S6_joint_cfm_s0s1_tau0	9.1982	28.0774	3.2511	0.8787

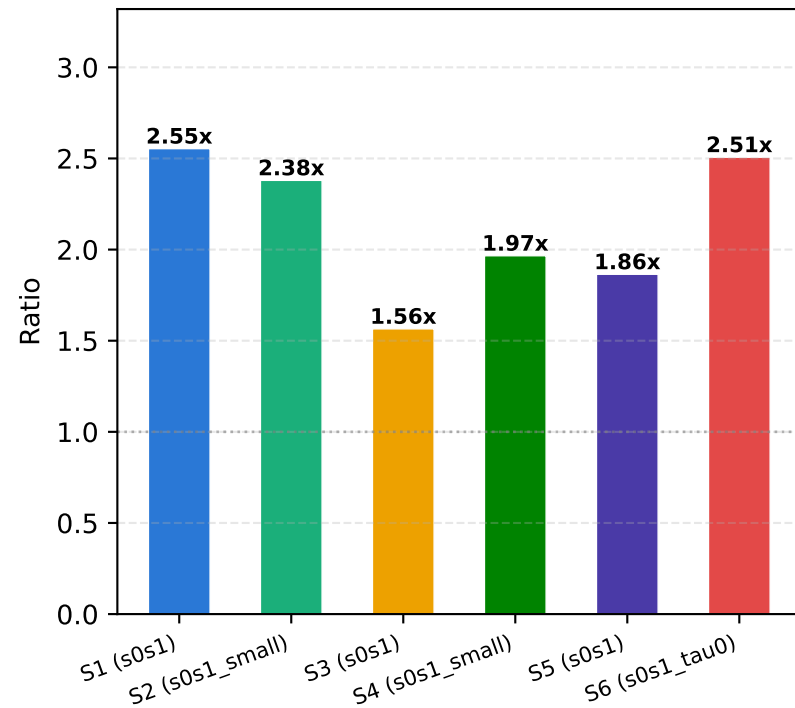
S0 Mean RMSE



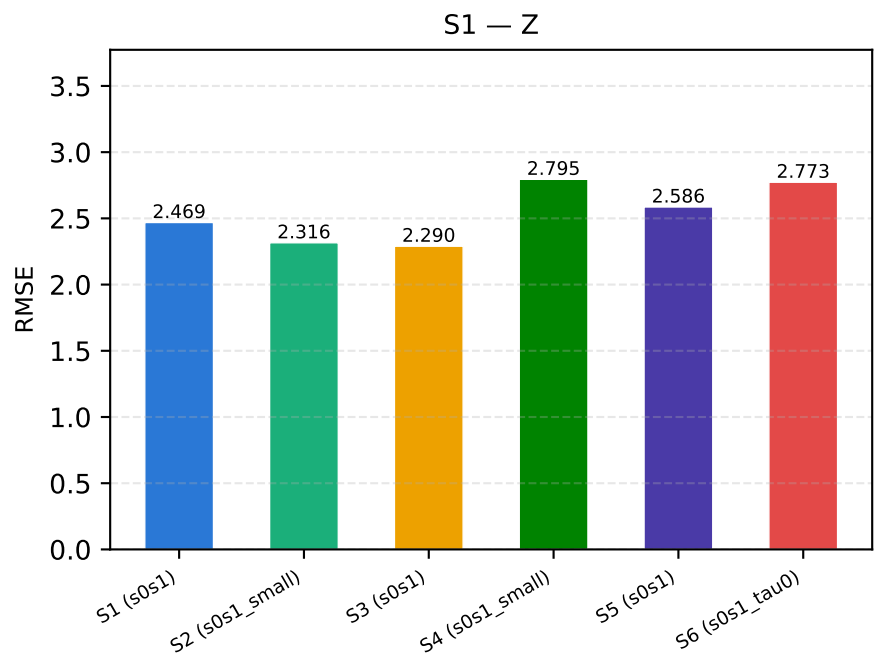
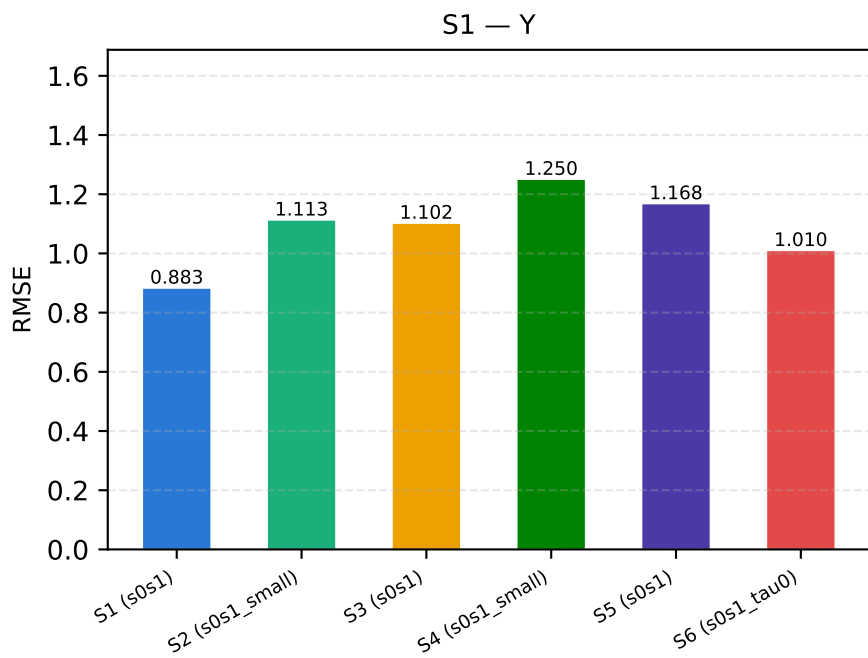
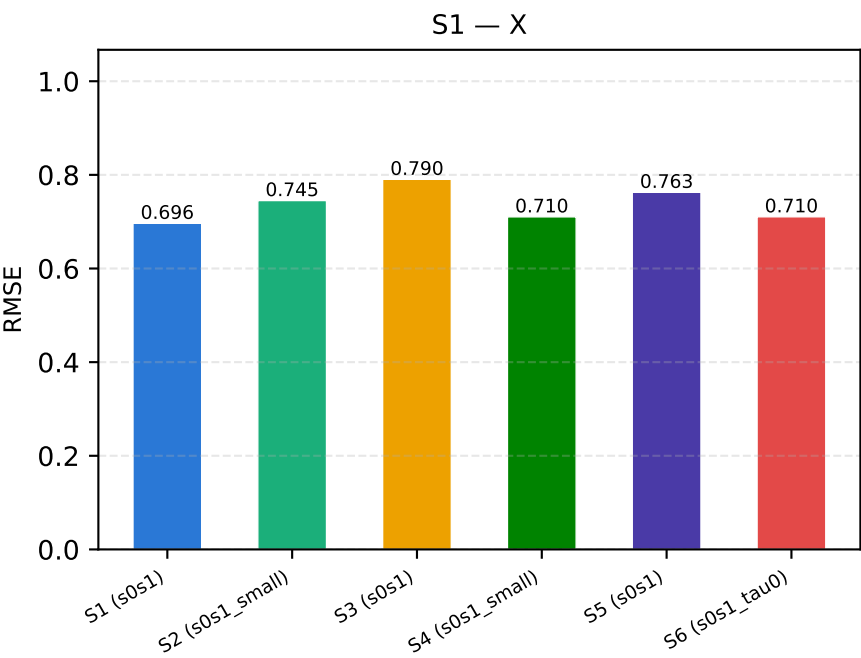
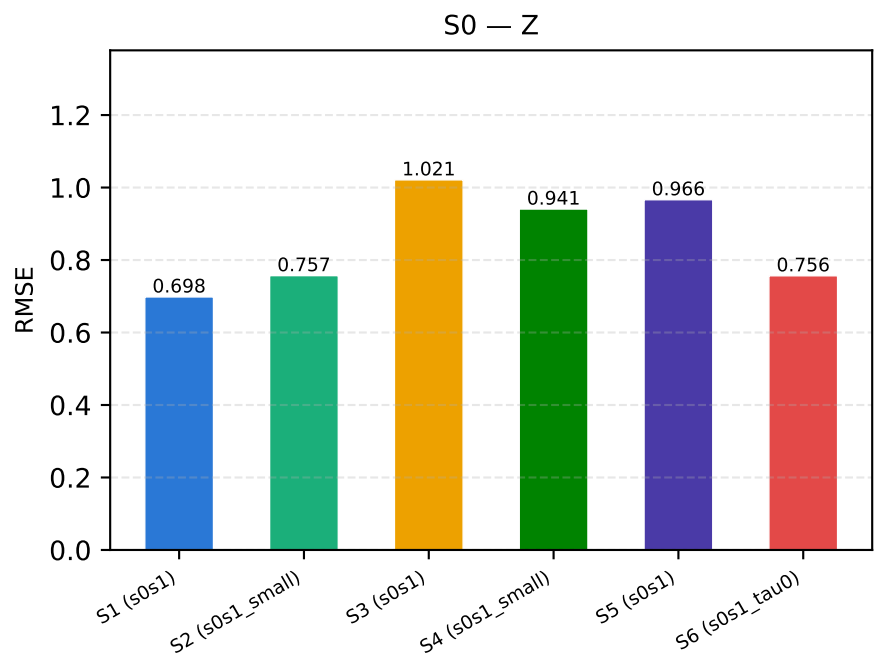
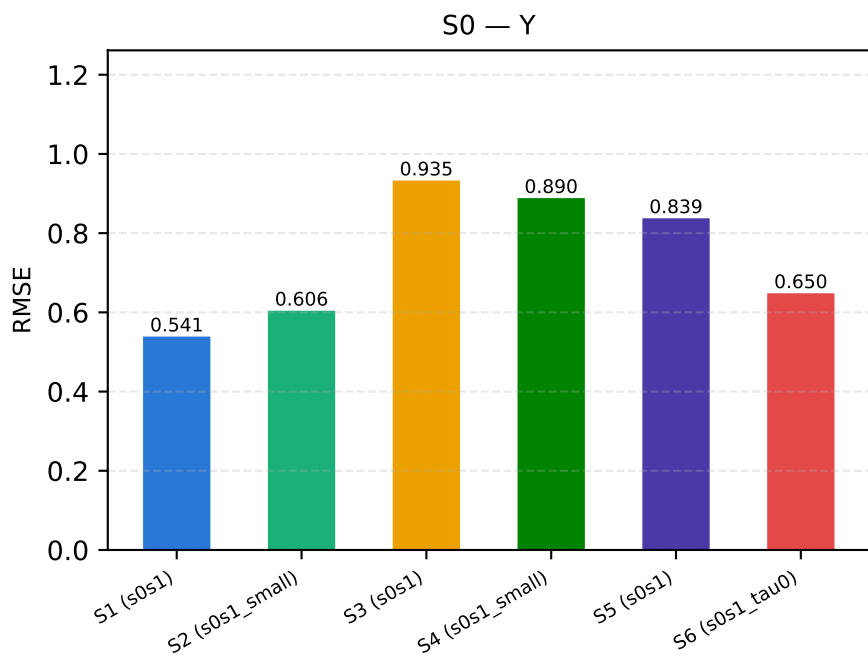
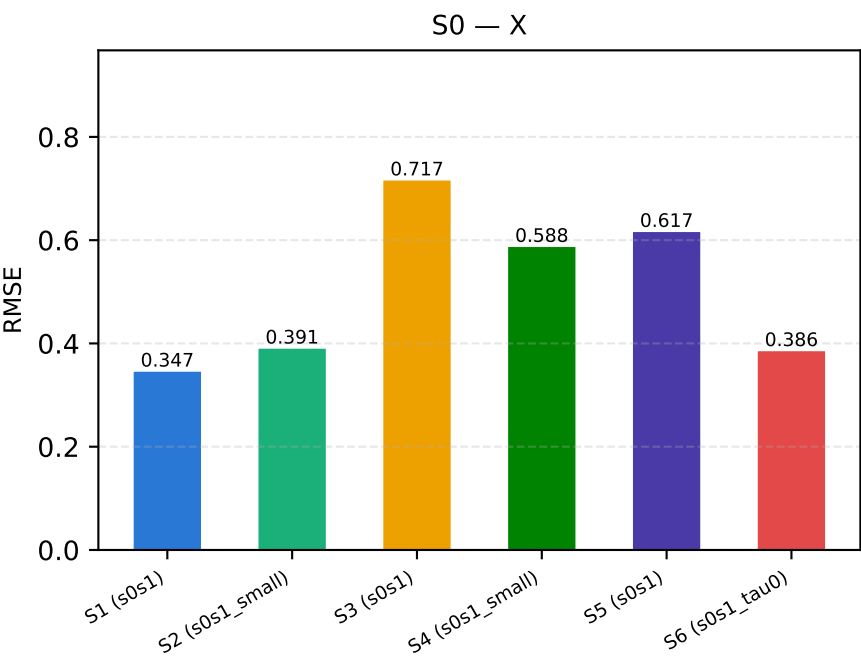
S1 Mean RMSE



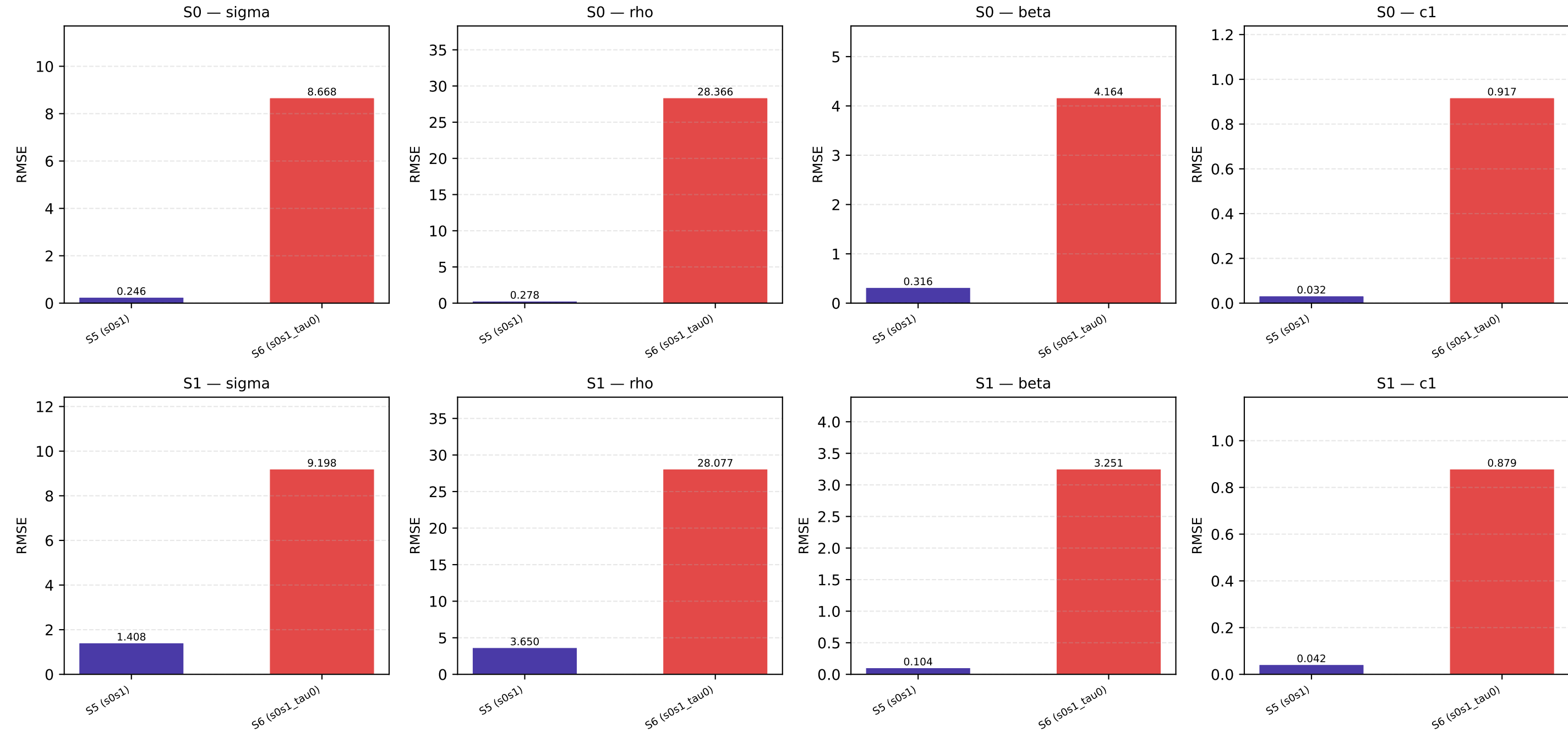
Degradation (S1/S0)



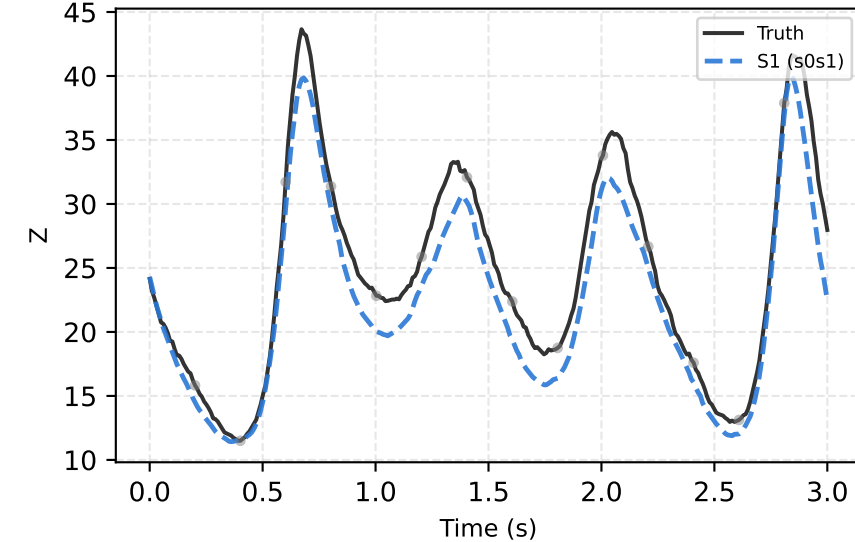
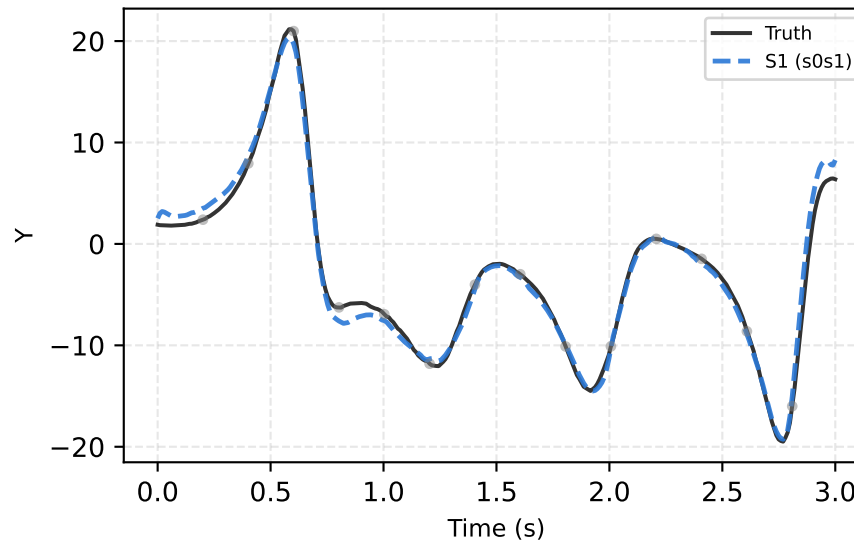
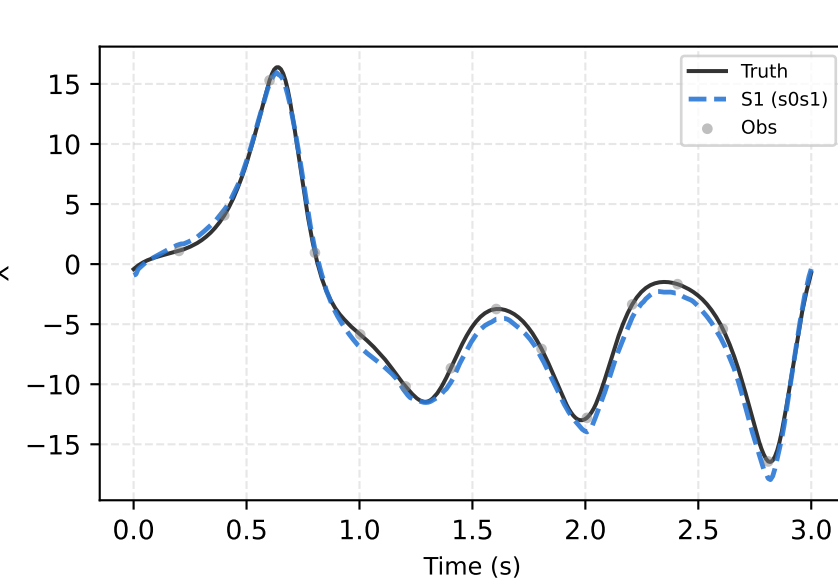
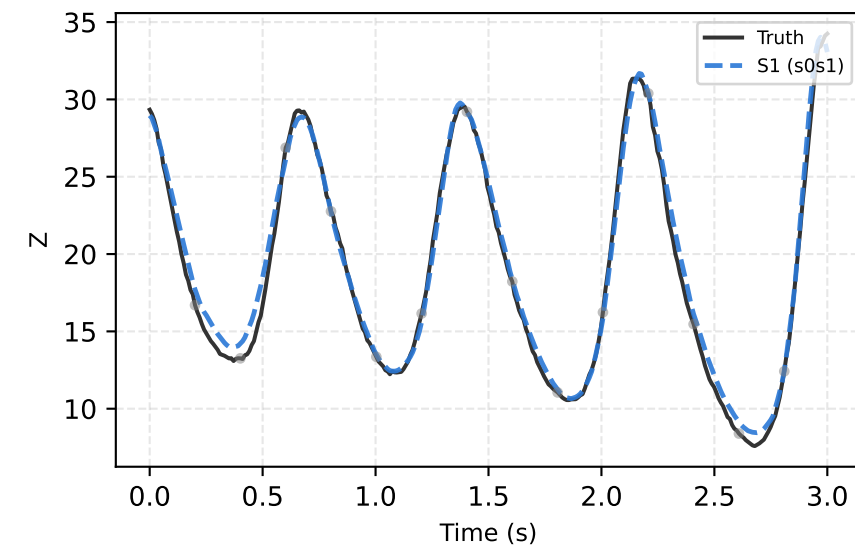
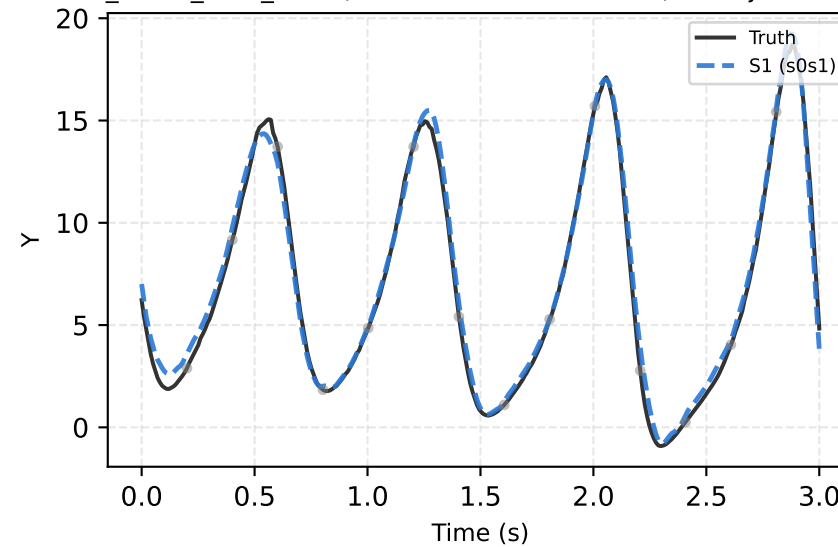
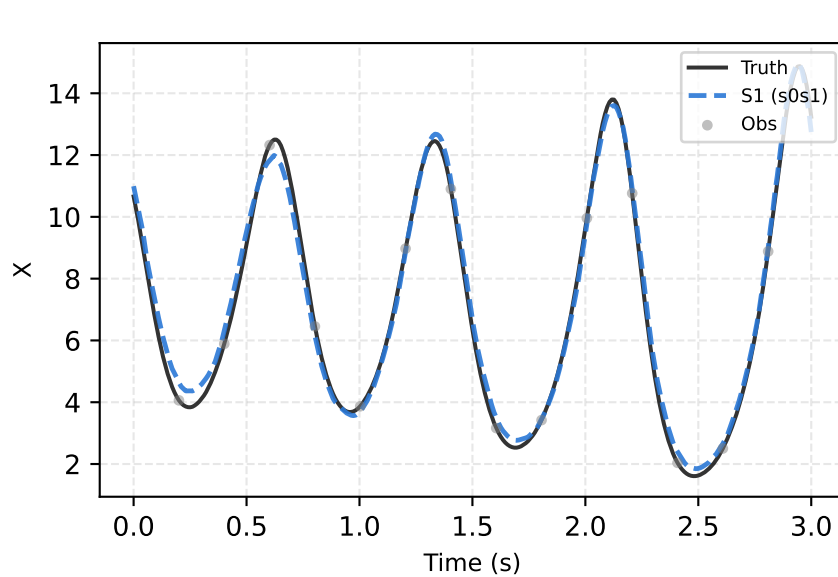
Per-Component RMSE Breakdown (S0, S1)



Parameter RMSE Breakdown (S0, S1) — JointCFM



Example Reconstructions — Best Model per Case (median-RMSE trajectory)



Notes

Notes

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All 6 experiments have results.